



Solution Submission by

VYOMAI



Because, Vyom is the limit...
We believe in using the vast expanse
of data to drive change



MEET THE TEAM



Vanshika Kaushik

Department of Computer Science,
IGDTUW

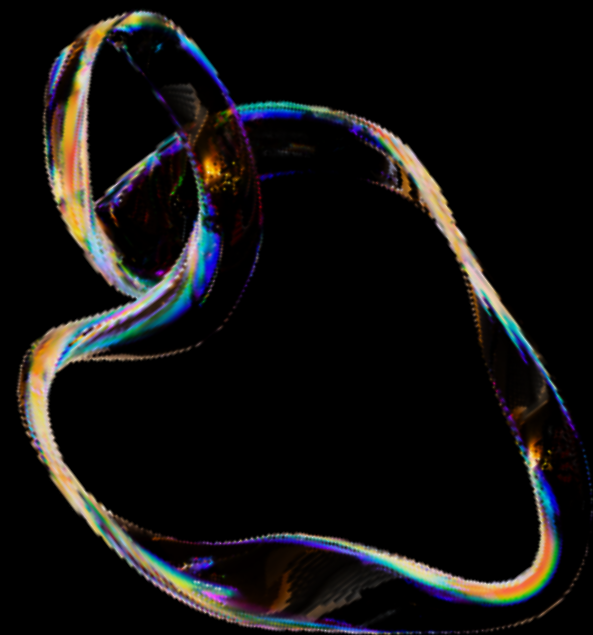
- Works on AI + Blockchain system integration
- Focused on building end-to-end intelligent decision systems



Urja Kohli

Department of Mechanical &
Automation Engineering, IGDTUW

- Works on AI/ML model development and training
- Focused on data-driven modeling and system optimization





PROBLEM STATEMENT

*When crops fail,
support arrives too late.*

Farmers already face climate uncertainty,
delayed insurance verification makes
recovery even harder.

Crop Data Already Exists

- Satellite imagery
- Weather data
- Crop health indicators
- PMFBY infrastructure



The Real Challenge

- Manual verification workflows
- Fragmented authority coordination
- Limited transparency in claim processing
- Delayed financial decision-making

Ecosystem Impact

- Slow payouts increase financial stress
- Reduced trust in insurance systems
- Poor coordination between stakeholders
- Weak climate resilience for farmers

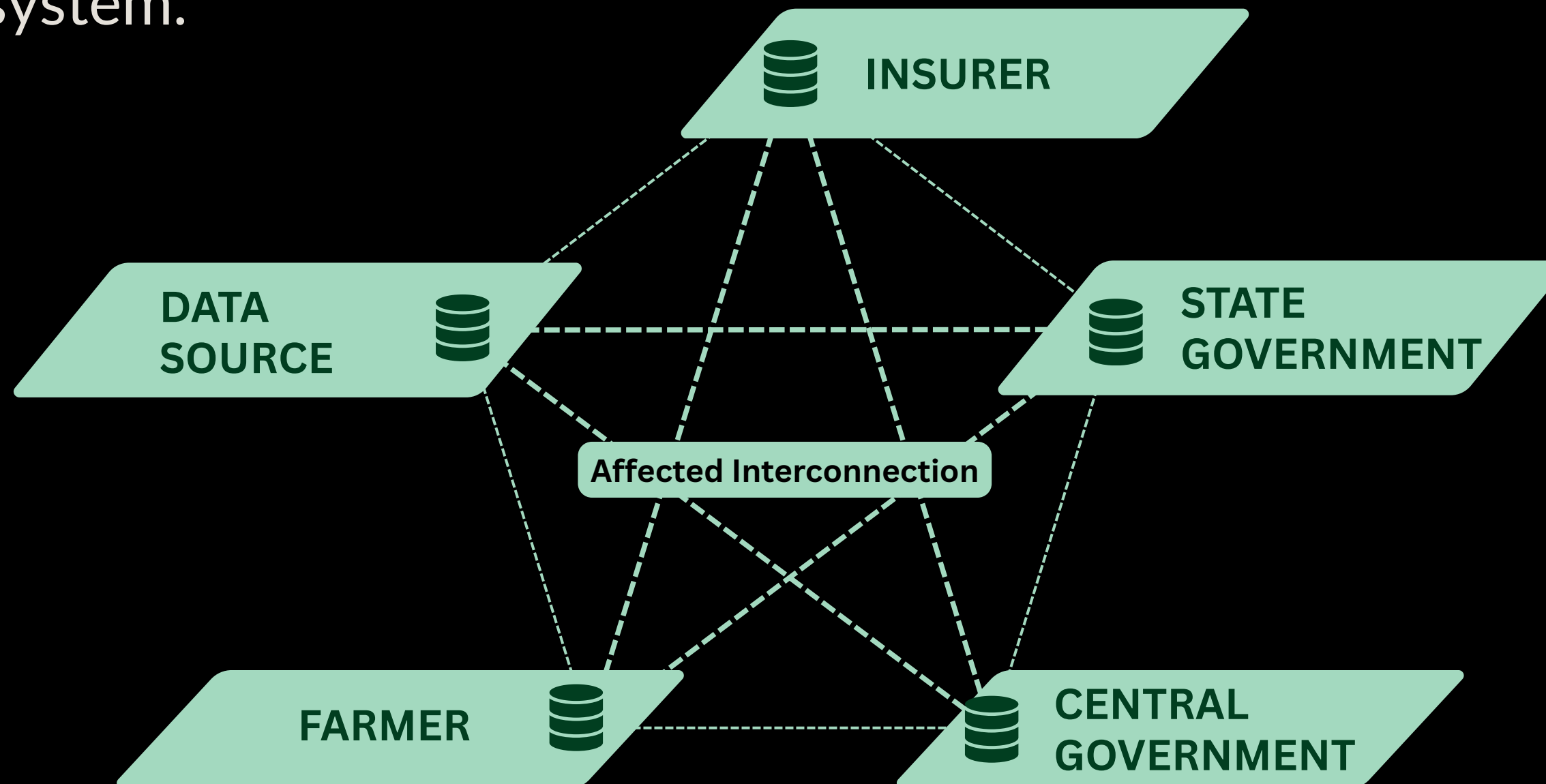
The challenge is not detecting crop loss,
it is creating a trusted, timely response system.

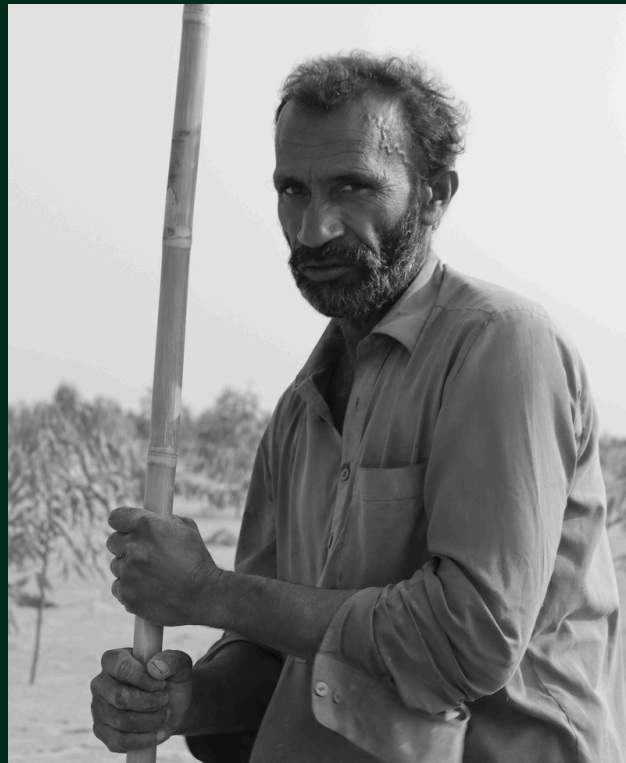


Interconnected Ecosystem



Climate uncertainty and delayed insurance workflows impact multiple interconnected stakeholders across the agricultural ecosystem.





Ramesh Kumar

- 35 years old
- Karnal, Haryana
- Small Farmer (2 acres)
- Depends on seasonal crops

USER PERSONA Farmer



ABOUT THE USER

Ramesh is a small farmer whose income depends entirely on seasonal crops. He faces frequent crop losses due to unpredictable weather and relies on insurance for financial support.

PROBLEMS

- Crop damage due to climate changes
- Insurance payouts take too long
- No clear information about claim status
- Threat to livelihoods and lives

CHALLENGES

- Managing expenses after crop loss
- Waiting weeks/months for compensation
- Lack of trust in current system

PERSONALITY

- Practical
- Hardworking
- Risk-aware
- Needs clarity

GOALS AND NEEDS

- Quick and reliable payouts
- Clear and simple updates
- Financial stability during losses
- To attain the help of the government



District Admin System

- Agriculture Dept
- Revenue / Land Records Dept
- Insurance Coordination Unit

USER PERSONA Government



ABOUT THE USER

Government authorities manage crop insurance across multiple departments, including land verification, crop assessment, and policy execution. They must coordinate data and decisions across systems.

PERSONALITY

- Accountability-driven
- System-oriented
- Efficiency-focused
- Risk-conscious

PROBLEMS

- Data scattered across departments
- Manual verification is slow
- Lack of unified system

GOALS AND NEEDS

- Unified decision system
- Faster verification and execution
- Auditable and transparent processes

CHALLENGES

- Cross-department coordination
- Delayed decision-making
- Ensuring transparency and fairness



Insurance Provider

Public/Private

- Works with govt schemes
- Handles claims and payouts

USER PERSONA Insurance Provider



ABOUT THE USER

The insurance provider processes claims and ensures payouts. They rely heavily on external verification and face challenges in ensuring accuracy and efficiency.

PERSONALITY

- Data-driven
- Risk-sensitive
- Efficiency-focused
- Compliance-oriented

PROBLEMS

- Fraudulent or disputed claims
- Delayed verification from authorities
- High operational workload

GOALS AND NEEDS

- Accurate claim validation
- Faster payout processing
- Reduced fraud and disputes

CHALLENGES

- Lack of real-time data
- Dependence on manual systems
- Difficulty in scaling operations



AI4India

Multiple stakeholders.

One **broken** system.

One **unified** solution.



Copy It from your
IDEA Submission



Solution Details

KrishiSetu

Small and marginal farmers often face delayed insurance support due to fragmented and manual verification systems during climate-related crop loss events.

KrishiSetu AI is an AI-driven agricultural insurance coordination platform that connects stakeholders, environmental data, and verification workflows into a unified response ecosystem.

Unlike traditional systems that rely heavily on slow manual processing and disconnected authorities, the platform supports faster, more transparent, and better-coordinated claim validation workflows.

The solution is designed for climate-vulnerable agricultural communities, insurance providers, verification authorities, and rural ecosystems dependent on seasonal farming.

⚡ Primary Differentiation

- Unified stakeholder ecosystem
- AI-assisted verification workflows
- Transparent claim coordination
- Climate-risk focused system design

🌍 Demographic It Impacts

- Small & marginal farmers
- Climate-vulnerable rural communities
- Insurance verification authorities
- Agricultural insurance ecosystems





Value proposition



Gain Creators

- Faster claim processing
- Improved coordination between authorities
- Better transparency & trust
- More reliable insurance workflows

Gains

- Timely financial support
- Clearer claim processing
- Increased trust in the system
- Better climate resilience support

■ exists in Current State
■ exists in Future State

Products & Services

- AI-assisted claim coordination platform
- Environmental & satellite data integration
- Unified verification workflow prototype
- Cross-department integration ecosystem
- Real-time nationwide deployment scalability



Customer Jobs

- Report crop loss
- Receive timely claim support
- Verify claims efficiently
- Coordinate across insurance workflows

Pain Relievers

- Reduces manual verification delays
- Minimizes fragmented communication
- Simplifies claim tracking & validation
- Supports quicker response during crop loss

Pains

- Delayed insurance payouts
- Lack of transparency
- Fragmented authority coordination
- Financial stress after crop damage

We are moving from fragmented insurance workflows, to coordinated agricultural response



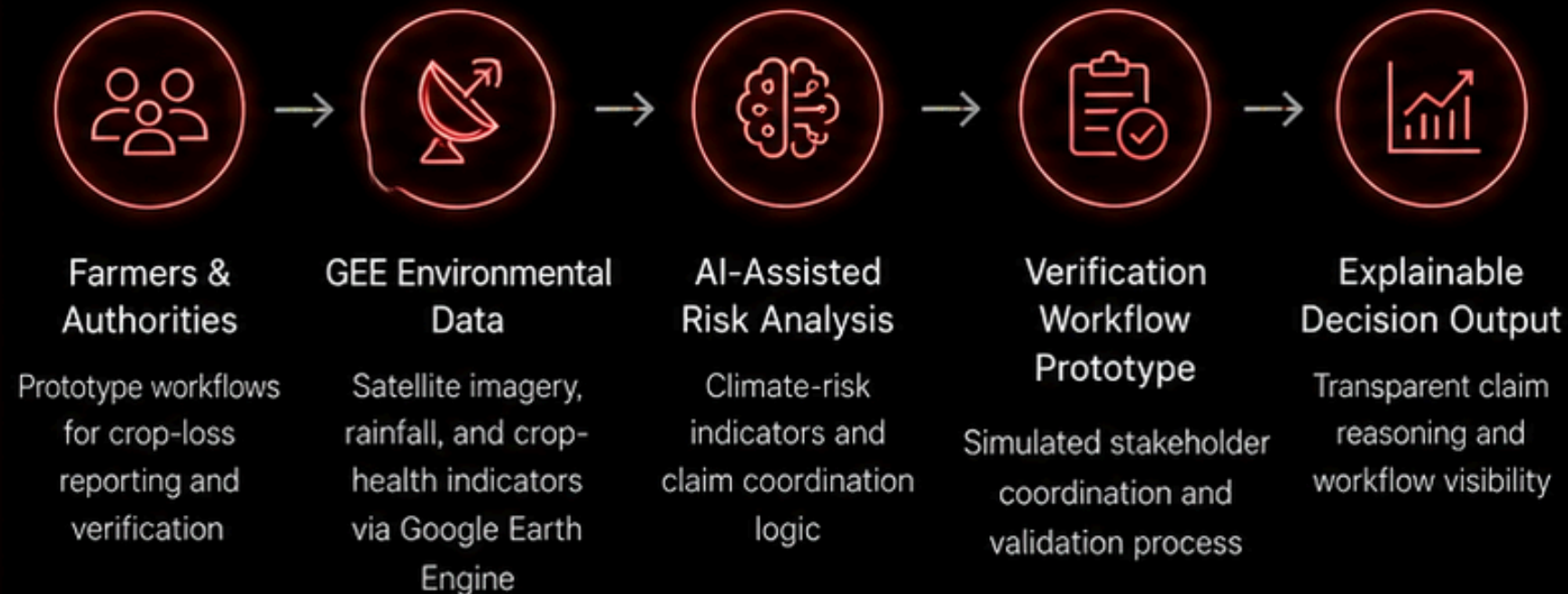
Thinking BIG

Current State vs Future State of our solution



CURRENT STATE – MVP

Prototype workflow with limited integration

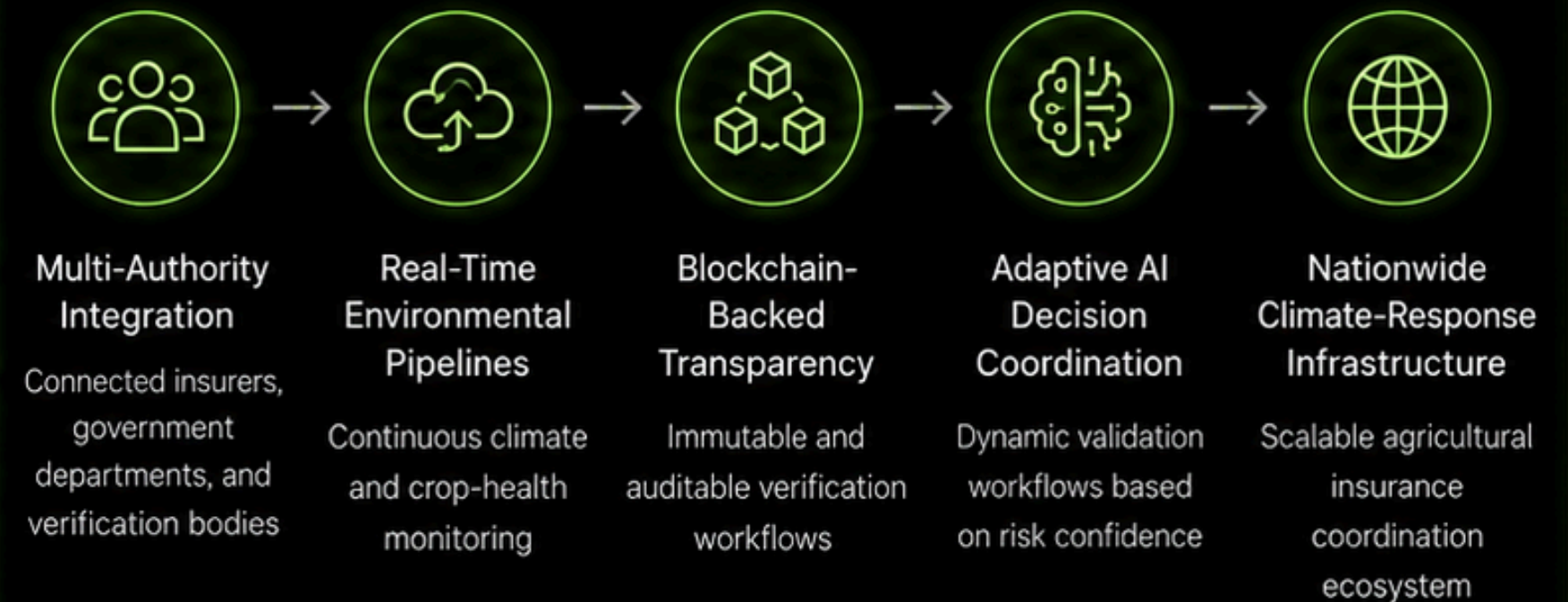


CURRENT LIMITATIONS

- ❌ Simulated authority integration
- ❌ Limited real-world datasets
- ❌ No live PMFBY integration
- ❌ Prototype-scale deployment only

FUTURE STATE – SCALABLE ECOSYSTEM

Integrated, intelligent and transparent agricultural insurance ecosystem



FUTURE ENABLERS

- ✅ Unified ecosystem coordination
- ✅ Faster & transparent claim workflows
- ✅ Scalable climate-risk response system
- ✅ Stronger trust across stakeholders



Architecture Discovery



Designed not just as a **product** — but as a **scalable agricultural response infrastructure**.

Climate Resilience

Faster Support

Trustworthy Verification

Ecosystem Coordination



AI4India



Assumptions



Technical Assumptions

- Environmental and satellite data are available through GEE and public climate datasets
- AI-assisted workflows support verification, not replacement of human authorities
- Modular architecture can scale across authorities and regions
- Blockchain used for transparency and auditability



Stakeholder Assumptions

- Farmers and authorities adopt digital workflows
- Insurers and departments coordinate via shared workflows
- Faster, transparent response is critical during climate-risk events



Data Assumptions

- Crop-health and weather indicators are meaningful for verification
- Public datasets are reliable and updated
- Off-chain storage securely manages large environmental datasets



Key Decisions



Technical Decisions

- Use Google Earth Engine (GEE) for satellite and environmental analysis
- AI-assisted climate-risk evaluation workflows
- Hybrid off-chain / on-chain architecture
- Explainability integrated into decision visibility



Ecosystem Decisions

- Farmers at the center of the ecosystem
- Support coordination between insurers, government and authorities
- Focus on transparency over full automation
- Build scalable, ecosystem-first architecture



Data Decisions

- Store heavy environmental data off-chain
- Store verification proof and logs on-chain
- Ensure data privacy, integrity and traceability
- Maintain minimal, meaningful on-chain data



Team Alignment

Shared vision, goals, priorities and collaboration.



Risk & Mitigation

Risk Category

Key Risks

Mitigation Strategies



Technical Risks

- AI prediction inaccuracies
- Incomplete / delayed datasets
- Scalability during high load
- Integration complexity

- ✓ Human-in-the-loop verification
- ✓ Multi-source data validation
- ✓ Modular, scalable architecture
- ✓ Phased integration approach



Stakeholder Risks

- Low trust in digital workflows
- Resistance to adoption
- Fragmented coordination

- ✓ Explainable decisions
- ✓ Transparent workflow visibility
- ✓ Unified coordination layer



Data Risks

- Data inconsistency
- Data privacy concerns
- Data accessibility issues

- ✓ Data standardization
- ✓ Secure off-chain storage
- ✓ Role-based access control



Dependencies



Technical Dependencies

- Google Earth Engine datasets and availability
- Weather / Climate APIs and external data sources
- Cloud infrastructure and storage services
- Blockchain network (future implementation)
- Third-party services for notifications and integrations



Ecosystem Dependencies

- Participation from insurers and authorities
- Government policy and procedural support
- Availability of ground verification teams
- Adoption by farmers and rural communities

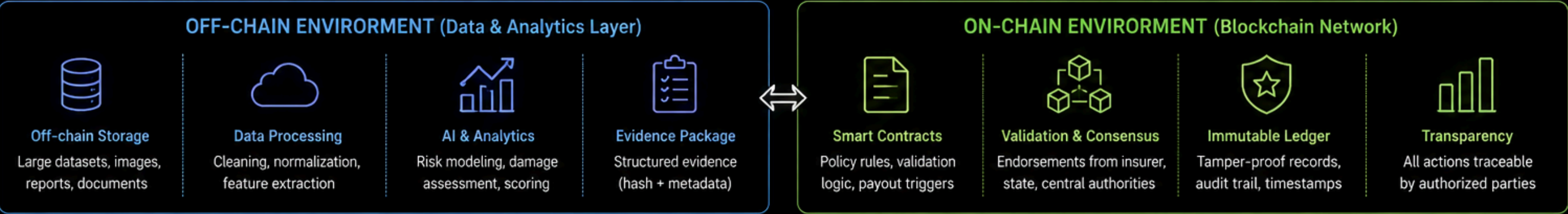
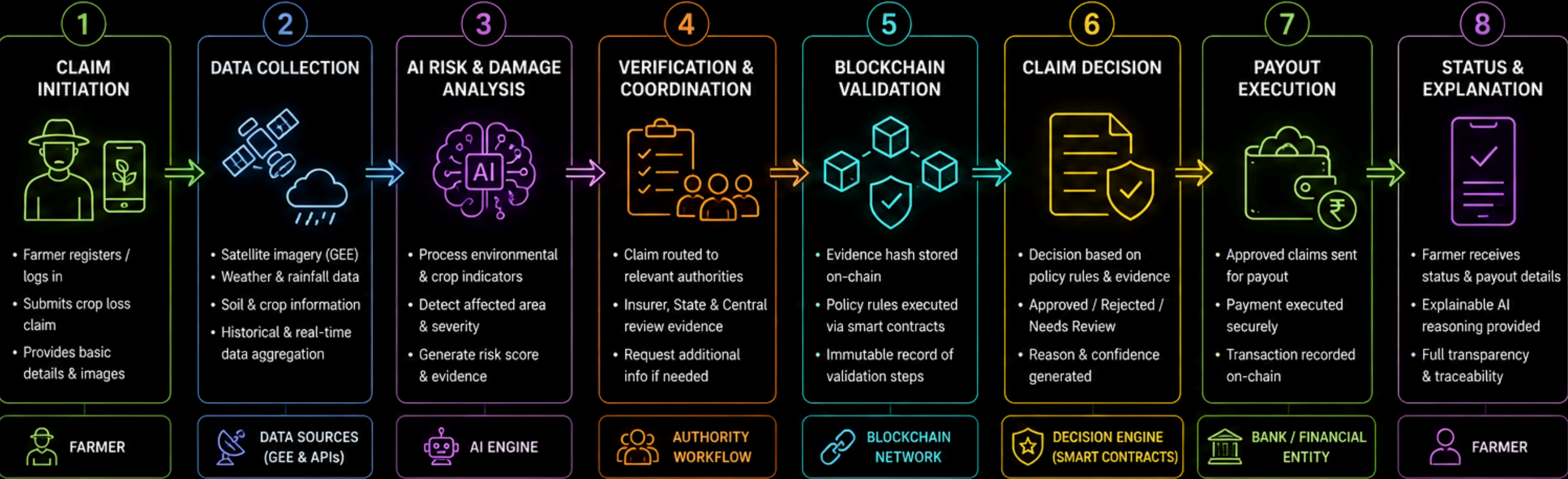


Data Dependencies

- Reliable satellite and environmental data
- Timely updates of crop and climate data
- Standardized data formats and protocols
- Data sharing agreements and permissions

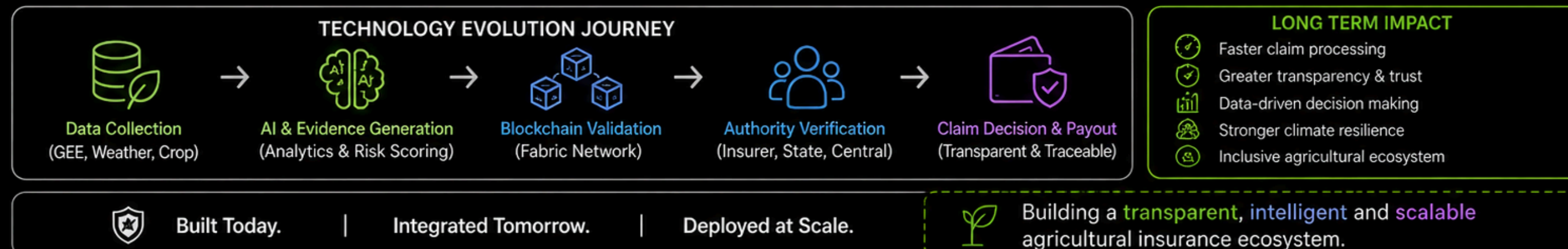
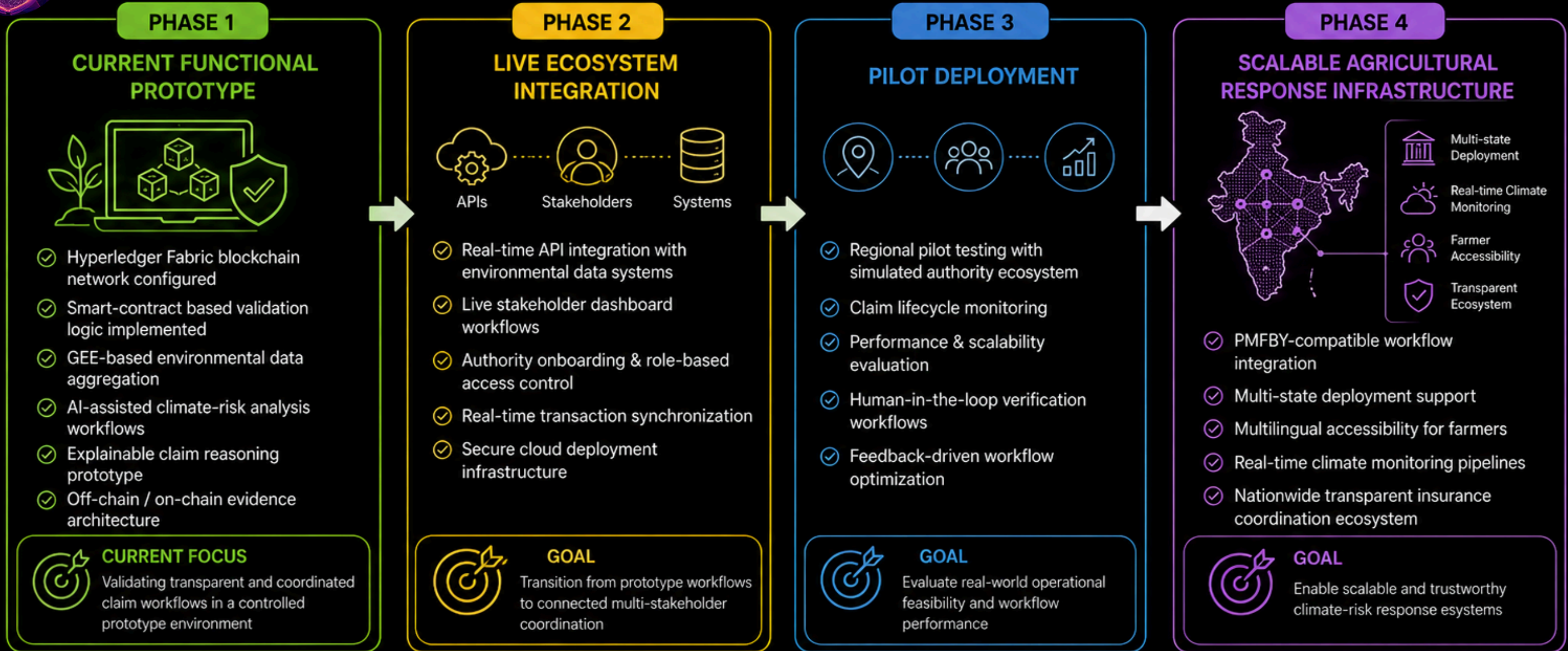


Process & Data Flow





Rollout Roadmap





Solution Inclusivity, Fairness and Explainability (1/2)



We are building an **inclusive**, **fair** and **transparent** AI-powered coordination system that prioritizes accessibility, representation and **human-centered design**.

1

Is the solution accessible to all users irrespective of language, digital access and disability?



Language Accessibility

- ✓ English & Hindi interface support in current prototype
- ✓ Designed for future multilingual regional language expansion



Digital Accessibility

- ✓ Mobile-first farmer interface
- ✓ Lightweight workflows for low-bandwidth rural regions
- ✓ Simple and guided interaction design



Accessibility Considerations

- ✓ Clear visual workflows and explainable outputs
- ✓ Simplified status visibility for non-technical users
- ✓ Human-assisted verification workflows supported



YES

We are committed to making the solution inclusive, accessible and usable for every farmer and stakeholder.

2

Is the data representative of all relevant user groups your solution targets?



PARTIALLY — CURRENTLY IN PROTOTYPE STAGE

- ✓ Current system primarily utilizes environmental and satellite datasets
- ✓ Designed to support diverse agricultural and climate-vulnerable regions
- ✓ Workflow logic is being structured to avoid region-specific bias
- ✓ Future deployment stages will require broader field validation datasets



CURRENT FOCUS

Building scalable and adaptable verification workflows rather than user-specific profiling models.

3

What demographics was your solution tested across?



PRIMARY FOCUS GROUPS

-  Small & marginal farmers
-  Climate-vulnerable agricultural regions
-  Rural agricultural ecosystems

GEOGRAPHIC SCOPE

-  Indian agricultural and climate conditions used for workflow simulation
- Environmental indicators derived from publicly available satellite datasets



IMPORTANT NOTE

Current testing is system-level and workflow-oriented.

Large-scale human deployment testing has not yet been conducted.



Solution Inclusivity, Fairness and Explainability (2/2)



KrishiSetu AI is designed as a **transparent**, **human-supervised**, and **ecosystem-aware** coordination system — not a fully automated black-box decision engine.



4

Does your data represent all relevant groups you are delivering the solution to?



- ✓ Environmental datasets are geographically broad and publicly sourced.
- ✓ Current workflows are designed to remain adaptable across regions and crop conditions.
- ✓ Future pilot deployments will improve regional and demographic representation.
- ✓ Continuous monitoring and validation will be required during scaling.



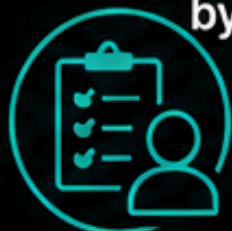
CURRENT APPROACH

Focus on infrastructure fairness, workflow transparency, and explainable coordination.



5

Can your solution be completely explained and has been validated by a Human?



- ✓ Explainable AI outputs provide transparent reasoning for claim decisions.
- ✓ Risk confidence scores and evidence summaries are generated.
- ✓ Human authorities remain part of the validation workflow.
- ✓ Human-in-the-loop review supported for uncertain or high-risk cases.
- ✓ Blockchain audit trails improve transparency and accountability.



EXPLAINABILITY FOCUS

Helping stakeholders understand why a claim decision was reached through clear, evidence-backed explanations.

6

Have you only collected the minimum necessary data for your solution?



- ✓ Primarily utilizes environmental and satellite datasets instead of sensitive personal profiling.
- ✓ Only essential farmer and claim-related information is required.
- ✓ Large environmental datasets are stored off-chain for efficiency.
- ✓ Blockchain stores verification metadata and audit records rather than sensitive raw data.
- ✓ Designed with data minimization and workflow transparency principles.



DATA MINIMIZATION

Collect only what is essential, secure it responsibly, and use it solely for building a transparent and trustworthy coordination system.

OFF-CHAIN



Large Environmental Data & Evidence

ON-CHAIN



Verification Metadata & Audit Records



Actual Solution Images/ Videos

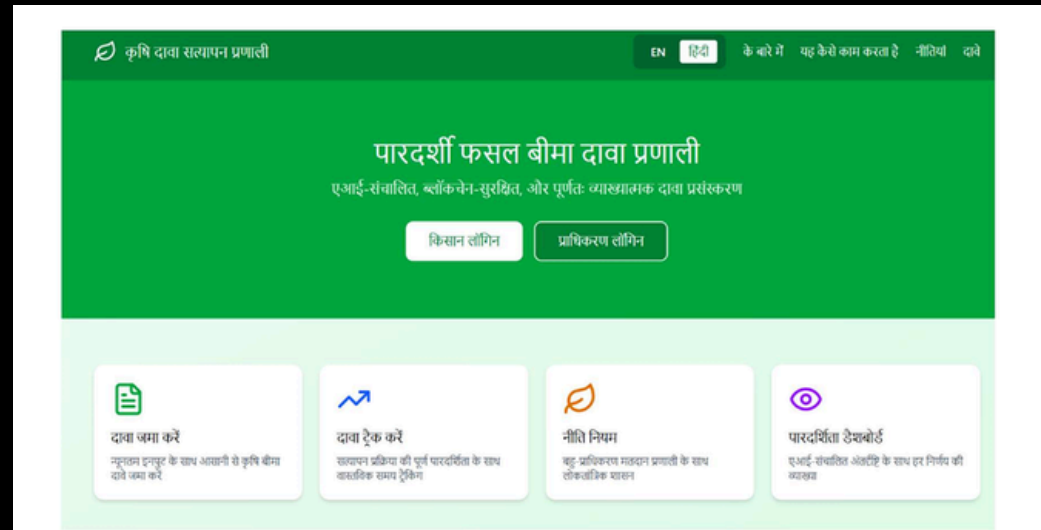


Figure 4. 6 Farmer Login Dashboard

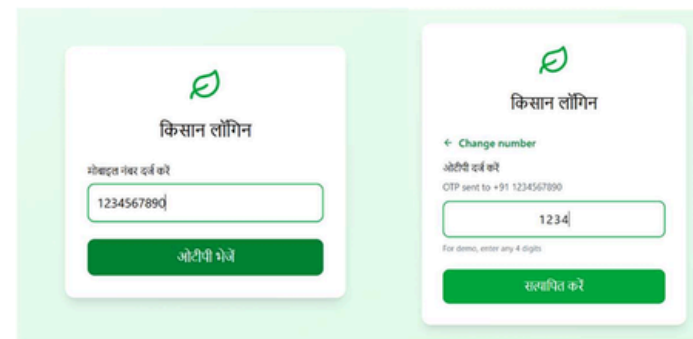


Figure 4. 7 Seamless Mobile/AADHAR verification login

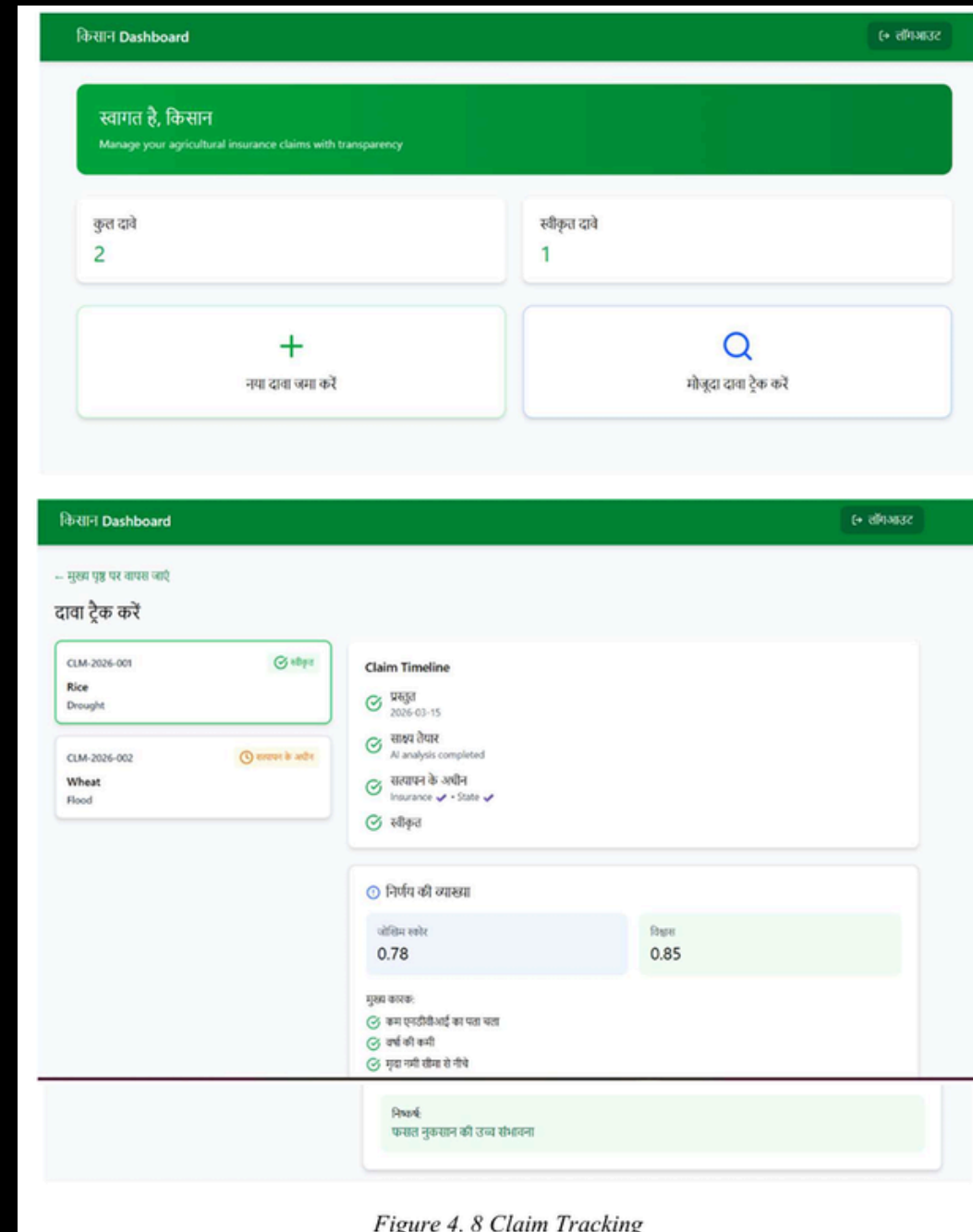


Figure 4. 8 Claim Tracking

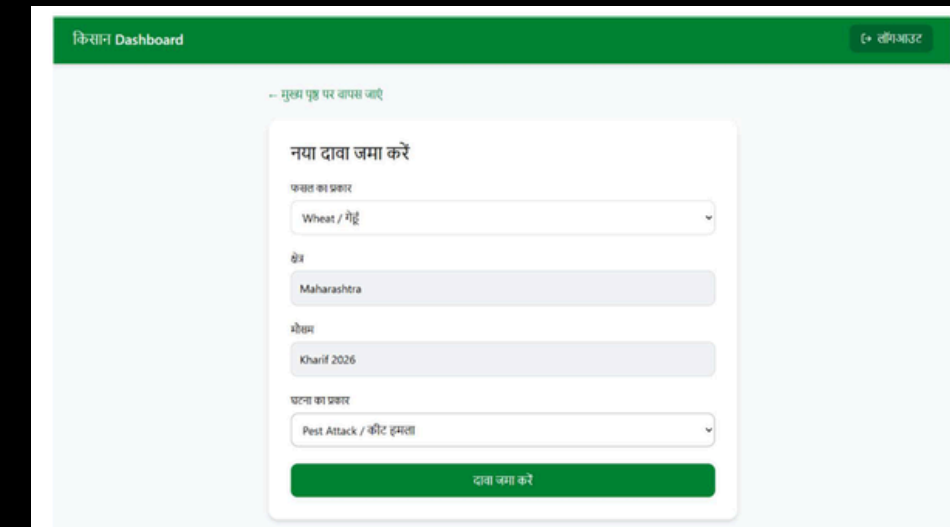


Figure 4. 9 Applying for Claim



Figure 4. 10 Retrieval of Geo-Information

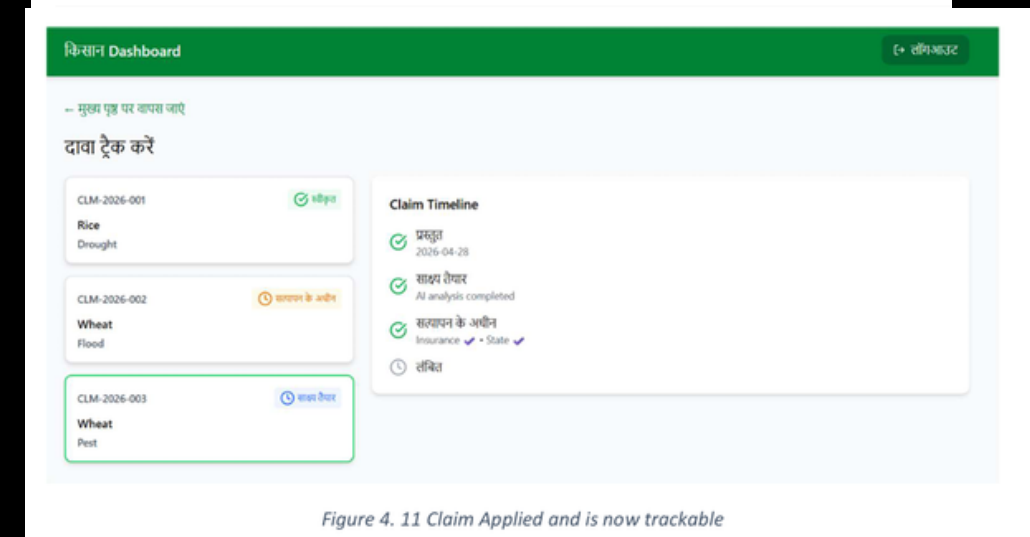


Figure 4. 11 Claim Applied and is now trackable

Live integrated walkthrough demo is currently being migrated as part of ongoing prototype environment reconfiguration.



Actual Solution Images/ Videos



Demo Media

Supporting Assets: Research Paper accepted and presented under the IEEE Conference for technical soundness

Live integrated walkthrough demo is currently being migrated as part of ongoing prototype environment reconfiguration.



THANK YOU

And all the best for
your solution judging

